spherical mirror. After reflection it retraces its path.

Reason: Angle of incidence (*i*) = Angle of reflection (*r*) = 0°.

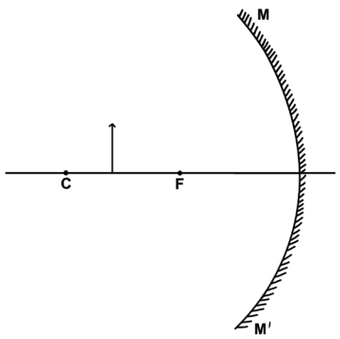
VI. Answer very briefly.

1. According to cartesion sign convention, which mirror and which lens has negative focal length?
2. Name the mirror(s) that can give (i) an erect and enlarged image, (ii) same sized, inverted image.
3. If an object is placed at the focus of a concave mirror, where is the image formed?
4. Why does a ray of light bend when it travels from one medium to another?
5. What is the speed of light in vacuum?
6. Concave mirrors are used by dentists to examine teeth. Why?

VII. Answer briefly.

1. a) Complete the diagram to show how a concave mirror forms the image of the object.

   b) What is the nature of the image?



2. Pick out the concave and convex mirrors from the following and tabulate them.

   Rear-view mirror, Dentist's mirror, Torch-light mirror, Mirrors in shopping malls, Make-up mirror.

3. State the direction of incident ray which after reflection from a spherical mirror retraces its path. Give reason for your answer.
4. What is meant by magnification? Write its expression. What is its sign for real image and virtual image?
5. Write the spherical mirror formula and explain the meaning of each symbol used in it.

VIII. Answer in detail.

1. a) Draw ray diagrams to show how the image is formed using a concave mirror, when the position of object is: i) at C ii) between C and F iii) between F and P of the mirror.

   b) Mention the position and nature of image in each case.

2. Explain with diagrams how refraction of incident light takes place from

   a) rarer to denser medium b) denser to rarer medium c) normal to the surface separating the two media.

IX. Numerical problems.

1. A concave mirror produces three times magnified real image of an object placed at 7 cm in front of it. Where is the image located? (Ans: 21 cm in front of the mirror)
2. Light enters from air into a glass plate having refractive index 1.5. What is the speed of light in glass? (Ans: 2×10^8 ms^{-1})
3. The speed of light in water is 2.25×10^8 ms^{-1}. If the speed of light in vacuum is 3×10^8 ms^{-1}, calculate the refractive index of water. (Ans: 1.33)

X. Higher Order Thinking Skills.

1. Light ray emerges from water into air. Draw a ray diagram indicating the change in its path in water.
2. When a ray of light passes from air into glass, is the angle of refraction greater than or less than the angle of incidence?
3. What do you conclude about the speed of light in diamond. if the refractive index of diamond is 2.41?

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